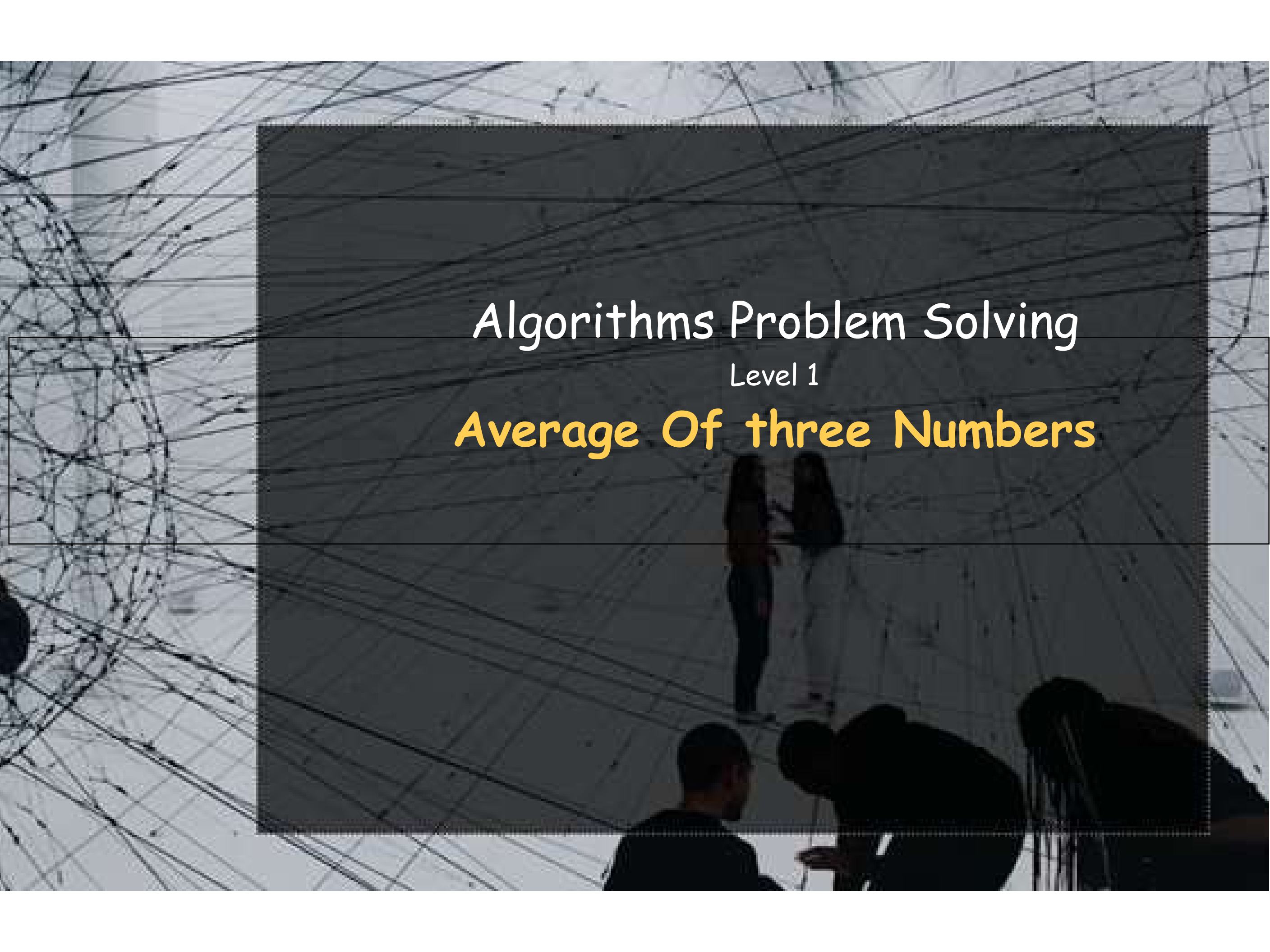


Algorithms Problem Solving

Level 1

Average Of Three Numbers



Algorithms Problem Solving

Level 1

Average Of three Numbers

Definition

- ◆ This program reads three numbers from the user and calculates their sum and average.
- ◆ The program uses a structure to group the three numbers and separate functions to read, calculate, and print the results.
- ◆ The average is calculated by dividing the sum of the three numbers by 3.

Explanation

- ◆ The struct `stNumbers` groups three integer values:
 - Num1 stores the first number.
 - Num2 stores the second number.
 - Num3 stores the third number.
- ◆ `ReadNumbers()` reads the three numbers from the user and returns an `stNumbers` object.
- ◆ `sumNumbers()` calculates and returns the sum of the three numbers.
- ◆ `avgNumbers()` calls `sumNumbers()` and divides the result by 3.
- ◆ The expression `(float)sumNumbers(numbers)` converts the sum to float before division so that a decimal average can be returned.
- ◆ `PrintResult()` prints both the sum and the average.
- ◆ In `main()`, the structure returned by `ReadNumbers()` is passed directly to `PrintResult()`:
`PrintResult(ReadNumbers());`

Steps in main()

1. Create the stNumbers structure.
2. Read the three numbers from the user.
3. Return the structure containing the numbers.
4. Pass the structure to PrintResult().
5. Call sumNumbers() to calculate the sum.
6. Call avgNumbers() to calculate the average.
7. Convert the sum to float before division.
8. Print the sum and average.

💡 Example

◆ If the user enters:

First number: 10

Second number: 20

Third number: 30

◆ The sum is:

$$10 + 20 + 30 = 60$$

◆ The average is:

$$60 / 3 = 20$$

➡ Result

The sum of the three numbers is: 60

The average of the three numbers is: 20

Note

- ◆ The function `avgNumbers()` returns float because the average can contain a decimal value.
- ◆ The expression `(float)sumNumbers(numbers)` ensures floating-point division.
- ◆ For example, if the sum is 61, the average is 20.3333 instead of 20.

⊘ Warning

- ◆ If the sum is divided as an integer, the decimal part can be lost.
- ◆ Converting the sum to float before dividing by 3 preserves the decimal result.



Comparison

▨ Integer Division

- ✓ Both operands are integers.
- ✓ The decimal part is removed.
- ✓ Example: $61 / 3 \rightarrow 20$.

▤ Floating-Point Division

- ✓ At least one operand is a floating-point value.
- ✓ The decimal part is preserved.
- ✓ Example: $(\text{float})61 / 3 \rightarrow 20.3333$.

✦✦ Summary

- ✓ A structure can group multiple related numbers into one object.
- ✓ The average is calculated by dividing the sum by 3.
- ✓ Converting the sum to float preserves decimal values in the average.

Thank you 😊